

Self assessment upper limb

PECTORAL REGION, BACK, SHOULDER, AXILLA & BRACHIAL PLEXUS

M.C.Q'

- 1- The suspensory ligament of axilla, mark correct statement:**
 - a- Continuation of the deep fascia of the neck.
 - b- Continuation of axillary sheath.
 - c- Continuation of clavipectoral fascia.
 - d- Continuation of suprapleural membrane.
- 2- The clavipectoral fascia is pierced by, mark the appropriate:**
 - a- Lateral pectoral nerve.
 - b- Medial pectoral nerve.
 - c- Suprascapular nerve.
 - d- Upper subscapular nerve.
- 3- The following structures doesn't pierce the clavipectoral fascia:**
 - a- Lateral thoracic artery.
 - b- Lateral pectoral nerve.
 - c- Thoracoacromial artery.
 - d- Cephalic vein.
- 4- Regarding pectoralis major, mark one correct statement:**
 - a- Inserted into the lateral lip of the bicipital groove.
 - b- Receives nerve supply from posterior cord of brachial plexus.
 - c- Forms the medial wall of axilla.
 - d- Is a powerful lateral rotator of the shoulder.
- 5- The serratus anterior muscle:**
 - a- Arises by eight digitations from the lower eight ribs.
 - b- Is inserted into the costal aspect of the medial border of scapula.
 - c- Is innervated by thoracodorsal nerve.
 - d- Is a powerful extensor of the shoulder.
- 6- Winging of scapula follows paralysis of which muscle?**
 - a- Serratus anterior muscle.
 - b- Deltoid muscle.
 - c- Trapezius muscle.
 - d- Teres minor muscle.

- 7- Upward rotation of the scapula in raising the arm above head, due to contraction of; mark the correct answer:**
- a- Trapezius and latissimus dorsi muscles.
 - b- Serratus anterior and rhomboid muscles.
 - c- Trapezius and serratus anterior muscles.
 - d- Trapezius and teres minor muscles.
- 8- The following muscle is attached to lateral border of scapula:**
- a- The serratus anterior muscle.
 - b- The teres major muscle.
 - c- The levator scapulae muscle.
 - d- The rhomboid major muscle.
- 9- The acromion process of scapula, choose the correct answer:**
- a- Continuous with the medial end of the scapular spine.
 - b- Its lateral border is continuous with upper lip of crest of the spine.
 - c- Gives origin to anterior fibers of deltoid.
 - d- Gives attachment to the lower fibers of trapezius.
- 10- Which of following muscles is inserted into greater tuberosity?**
- a- Subscapularis muscle.
 - b- Teres minor muscle.
 - c- Teres major muscle.
 - d- Latissimus dorsi muscle.
- 11- The greater tuberosity of the humerus, choose correct answer:**
- a- Gives attachment to subscapularis.
 - b- Gives attachment to pectoralis major.
 - c- Gives attachment to teres minor.
 - d- Gives attachment to teres major.
- 12- The following muscles abducts the arm from 0-15°:**
- a- Upper and lower fibers of trapezius muscle.
 - b- Supraspinatus muscle.
 - c- Infraspinatus muscle.
 - d- Deltoid muscle.

- 13- The following muscles can extend the arm:**
- a- Pectoralis major muscle.
 - b- Teres major muscle.
 - c- Biceps brachii muscle.
 - d- Coracobrachialis muscle.
- 14- Which one of the following muscles that can extend the arm?**
- a- Infraspinatus.
 - b- Latissimus dorsi.
 - c- Clavicular fibers of pectoralis major.
 - d- Anterior fibers of deltoid.
- 15- The following muscle can abduct the arm:**
- a- Pectoralis major muscle.
 - b- Teres major muscle.
 - c- Supraspinatus.
 - d- Latissimus dorsi
- 16- One of the following muscle is a rotator cuff muscle:**
- a- Infraspinatus.
 - b- Pectoralis major.
 - c- Serratus anterior.
 - d- Teres major.
- 17- Deltoid muscle fibers are:**
- a- Parallel muscle fibers.
 - b- Unipennate muscle fibers.
 - c- Multipennate muscle fibers.
 - d- Circumpennate muscle fibers.
- 18- Regarding the deltoid muscle, mark one correct statement:**
- a- Its middle fibers adduct the arm.
 - b- Its posterior fibers flex and medially rotate the arm.
 - c- Its motor nerve arises from lateral cord of brachial plexus.
 - d- Its paralysis results in a flat shoulder.
- 19- The deltoid muscle, choose the correct answer:**
- a- Arises from clavicle, acromion and coracoid process.
 - b- Is inserted into deltoid tuberosity.
 - c- Is innervated by radial nerve.
 - d- Its middle fibers induce lateral rotation of scapula.

- 20- Which one of the following nerves innervates a muscle that abducts shoulder from 15 to 90 degrees?**
- a- Radial nerve.
 - b- Ulnar
 - c- Axillary nerve.
 - d- Suprascapular nerve.
- 21- Regarding quadrangular space in axillary region; one is correct:**
- a- It is bounded above by the supraspinatus.
 - b- It is bounded laterally by the surgical neck of the humerus.
 - c- It is bounded below by the teres minor.
 - d- It contains the circumflex scapular artery.
- 22- Structures passing through the quadrangular space of the axilla; mark the correct answer:**
- a- The axillary nerve and posterior circumflex humeral artery.
 - b- The axillary nerve and anterior circumflex humeral artery.
 - c- The axillary nerve and circumflex scapular artery.
 - d- The axillary nerve and musculocutaneous nerve.
- 23- Upper (medial) triangular space of axilla, choose correct answer:**
- a- Upper border by teres minor.
 - b- Lower border by subscapularis.
 - c- Transmits axillary nerve.
 - d- Transmits lower subscapular nerve.
- 24- Lower (lateral) triangular space of axilla, choose correct answer:**
- a- Upper border by teres minor.
 - b- Medial border by lateral head of triceps.
 - c- Transmits radial nerve and profound brachii artery.
 - d- Transmits median nerve and radial recurrent artery.
- 25- Regarding the axilla:**
- a- Posterior wall is formed of subscapularis, teres minor, latissimus dorsi.
 - b- It contains the roots of the brachial plexus.
 - c- Pectoralis major & minor share in the formation of its anterior wall.
 - d- The medial wall is the shortest wall.
- 26- The cervico-axillary canal, choose the correct answer:**
- a- Bounded medially by inner border first rib.
 - b- Bounded posteriorly by upper border of scapula.
 - c- The roots of brachial plexus pass through it.
 - d- Transmits subclavian vessels.

- 27- The axillary septum doesn't includes:**
- a-** Pectoralis minor muscle and fascia.
 - b-** Suspensory ligament of axilla.
 - c-** Axillary sheath.
 - d-** Subclavius muscle and fascia.
- 28- Which one of the following nerves arises from the roots of the brachial plexus?**
- a-** Medial pectoral nerve.
 - b-** Dorsal scapular nerve.
 - c-** Axillary nerve.
 - d-** Thoracodorsal nerve.
- 29- The upper trunk of brachial plexus gives origin to; select the correct answers:**
- a-** Suprascapular nerve.
 - b-** Long thoracic nerve.
 - c-** Upper subscapular nerve.
 - d-** Axillary nerve.
- 30- The suprascapular ligament lies between, mark correct answer:**
- a-** Suprascapular and subscapular arteries.
 - b-** Suprascapular nerve and suprascapular artery.
 - c-** Suprascapular and upper subscapular nerves.
 - d-** Upper and lower subscapular nerves.
- 31- Structures passing through the spinoglenoid notch, choose the most correct answer:**
- a-** Circumflex scapular artery.
 - b-** Suprascapular nerve and vessels.
 - c-** Subscapular nerve and vessels.
 - d-** Dorsal scapular vessels.
- 32- Which nerve arises from a cord of the brachial plexus cords?**
- a-** Long thoracic nerve.
 - b-** Nerve to subclavius.
 - c-** Suprascapular nerve.
 - d-** Thoracodorsal nerve.

33- Medial cord of brachial plexus gives, choose the correct answer:

- a- Lateral cutaneous nerve of the forearm.
- b- Musculocutaneous nerve.
- c- Lateral pectoral nerve.
- d- Ulnar nerve.

34- Posterior cord of brachial plexus gives, choose correct answer:

- a- Suprascapular nerve.
- b- Long thoracic nerve.
- c- Axillary nerve.
- d- Median nerve.

35- Which one is a branch of the lateral cord of the brachial plexus?

- a- Upper subscapular nerve.
- b- Long thoracic nerve.
- c- Musculocutaneous nerve.
- d- Suprascapular nerve.

36- Which muscles is supplied by medial cord of brachial plexus?

- a- Teres major.
- b- Deltoid.
- c- Anconeus.
- d- Pronator teres.

37- Following nerves are related directly to surgical neck of humerus:

- a- Radial nerve.
- b- Ulnar nerve.
- c- Axillary nerve.
- d- Musculocutaneous nerve.

38- Nerve commonly injured after fracture of humerus surgical neck is:

- a- Axillary nerve.
- b- Ulnar nerve.
- c- Radial nerve.
- d- Median nerve.

39- Which muscles are innervated by the axillary nerve?

- a- Supraspinatus and infraspinatus muscles.
- b- Deltoid and teres minor muscles.
- c- Pectoralis major and pectoralis minor muscles.
- d- Latissimus dorsi and teres major muscles.

- 40- The axillary nerve innervates; select the correct answer:**
- a-** Deltoid and supraspinatus.
 - b-** Deltoid and teres minor.
 - c-** Teres minor and teres major.
 - d-** Deltoid and teres major.
- 41- Inability to abduct arm from 15 to 90 degrees is due to injury of:**
- a-** Radial nerve.
 - b-** Long thoracic nerve (nerve to serratus anterior).
 - c-** Musculocutaneous nerve.
 - d-** Axillary nerve.
- 42- Regarding the axillary nerve:**
- a-** It arises from the lateral cord of the brachial plexus.
 - b-** It supplies the skin of the lower half of the lateral side of the arm.
 - c-** It passes through the triangular space.
 - d-** It supplies the deltoid and teres minor muscles.
- 43- The radial nerve consists of the following roots; choose the correct answers:**
- a-** Nerve roots C6, 7 & 8.
 - b-** Nerve root C6, 7 & 8.
 - c-** Nerve roots C5, 6, 7, 8c and T1.
 - d-** Nerve roots C5, 6, 7 and 8.

1- C	24- C
2- A	25- C
3- A	26- B
4- A	27- C
5- B	28- B
6- A	29- A
7- C	30- B
8- B	31- B
9- All wrong	32- D
10- B	33- D
11- C	34- C
12- B	35- C
13- B	36- D
14- B	37- C
15- C	38- A
16- A	39- B
17- C	40- B
18- D	41- D
19- B	42- D
20- C	43- C
21- B	
22- A	
23- A	

FILL IN THE BLANKS

- 1- The vein, the nerve and the..... artery pierce the clavipectoral fascia.
- 2- The trapezius receives motor fibers from and sensory fibers from
- 3- The anterior fibers of deltoid can do and rotation of the shoulder joint.
- 4- The posterior fibers of deltoid can do and rotation of the shoulder joint.
- 5- The middle fibers of deltoid abduct the arm from degrees to degrees.
- 6- The rotator cuff muscles include , , and
- 7- The nerve and the artery pass through the quadrangular space of the axilla.
- 8- The upper triangular space of the axilla transmits the vessels while the lower triangular space transmits the nerve and the vessels.
- 9- The cervicoaxillary canal is bounded by, and
- 10- The skin of the floor of the axilla is supplied by the nerve. Its root value is.....
- 11- The posterior wall of the axilla is formed of, and
- 12- The branches of the roots of the brachial plexus include and
- 13- The rhomboid muscles receive nerve supply from which arises from
- 14- The Supraspinatus and infraspinatus muscles are supplied by the nerve that arises from
- 15- Thepasses below the suprascapular ligament within the suprascapular foramen, while the passes above the ligament.
- 16- The branches of the trunks of the brachial plexus include and
- 17- The branches of the lateral cord of the brachial plexus are, and
- 18- The branches of the medial cord of the brachial plexus are, ,, and
- 19- The lateral cutaneous nerve of the forearm is a branch of, while the medial cutaneous nerve of the forearm is a branch of
- 20- The branches of the posterior cord of the brachial plexus are, ,, and

-
- 1- Cephalic, lateral pectoral, Thoracoacromial.**
 - 2- Spinal accessory nerve, C 3,4.**
 - 3- Flexion, medial.**
 - 4- Extension, lateral.**
 - 5- 15, 90.**
 - 6- Subscapularis, Supraspinatus, infraspinatus, teres minor.**
 - 7- Axillary, posterior circumflex humeral.**
 - 8- Circumflex scapular, radial, profunda brachii.**
 - 9- Outer border of 1st rib, middle 1/3 of clavicle, upper border of scapula.**
 - 10- The intercostobrachial, T2.**
 - 11- Subscapularis, teres major, latissimus dorsi.**
 - 12- Dorsal scapular nerve, long thoracic nerve.**
 - 13- Dorsal scapular nerve, C5 root of brachial plexus.**
 - 14- Suprascapular, upper trunk of the brachial plexus.**
 - 15- Suprascapular nerve, suprascapular artery.**
 - 16- Suprascapular nerve, nerve to subclavius.**
 - 17- Musculocutaneous nerve, lateral root of median nerve, lateral pectoral nerve.**
 - 18- Ulnar nerve, medial root of median nerve, medial pectoral nerve, medial cutaneous nerve of arm, medial cutaneous nerve of forearm.**
 - 19- Musculocutaneous nerve, medial cord of brachial plexus.**
 - 20- Radial nerve, axillary nerve, upper and lower subscapular nerves, thoracodorsal nerve.**

CROSS MATCHING

-1-

Regarding the muscles attached to the upper part of the humerus, choose from column (B) what suits the statements in column (A):

Column A	Column B
1- Subscapularis	a) Upper facet of greater tuberosity
2- Supraspinatus	b) Middle facet of greater tuberosity
3- Teres major	c) Lower facet of greater tuberosity
4- Teres minor	d) Lesser tuberosity
5- Infraspinatus	e) Medial lip of bicipital groove

-2-

Select from column (B) the action that suit the muscle in column (A):

Column A	Column B
1- Infraspinatus	a) Retraction of scapula
2- Rhomboid major	b) Flexion of arm
3- Serratus anterior	c) Adduction of arm
4- Latissimus dorsi	d) Protraction of scapula
5- Anterior fibers of deltoid	e) Lateral rotation of arm

-1-	-2-
1-d	1-e
2-a	2-a
3-e	3-d
4-c	4-c
5-b	5-b

PROBLEM SOLVING

-1-

A 45 years old woman was admitted to hospital with dislocation of her right shoulder.

- 1- What nerve is endangered by this dislocation?
- 2- What are the muscles supplied by this nerve?
- 3- What movements would be affected if this nerve was injured?
- 4- What would be the deformity resulting from this paralysis?
- 5- Where would be the skin sensory loss resulting from such an injury?
 - 1- Axillary nerve.
 - 2- Deltoid and teres minor.
 - 3- Loss of abduction of the shoulder from 15 to 90 degrees.
 - 4- Flat shoulder.
 - 5- Skin over the lower half of deltoid.

ARM, FRONT OF FOREARM, MEDIAN & ULNAR NERVES

M.C.Q'S

- 1- Lateral intermuscular septum, choose the correct answer:**
 - a- Extends between lesser tuberosity and lateral epicondyle.
 - b- Gives origin to brachioradialis.
 - c- Gives origin to extensor carpi radialis brevis.
 - d- Is pierced by radial nerve & branch of posterior interosseous artery.
- 2- The lateral intermuscular septum, choose the correct answer:**
 - a- Pierced by the ulnar nerve.
 - b- Gives origin to brachialis muscle.
 - c- Gives attachment to the lateral head of triceps.
 - d- Gives attachment to the supinator muscle.
- 3- The medial intermuscular septum; mark the correct answer:**
 - a- Pierced by the ulnar nerve.
 - b- Pierced by the profunda brachii arteries.
 - c- Gives attachment to the anconeus muscle.
 - d- Gives attachment to the lateral head of triceps muscle.
- 4- Regarding Biceps brachii, choose the correct answer:**
 - a- Its short head lies inside the capsule of the shoulder joint.
 - b- It flexes the elbow joints.
 - c- It pronates the forearm.
 - d- It is supplied by median nerve.
- 5- Which tendon is intracapsular extrasynovial of the shoulder joint, mark the correct answer:**
 - a- Long head of biceps brachii.
 - b- Short head of biceps brachii.
 - c- Long head of triceps muscle.
 - d- Teres minor.
- 6- Biceps brachii:**
 - a- It inserts in the ulnar tuberosity.
 - b- Is pierced by musculocutaneous nerve.
 - c- Is a powerful muscle in inducing supination.
 - d- Helps in extending the flexed elbow.

- 7- The coracoid process of the scapula gives attachment to:**
- a- Pectoralis minor.
 - b- Capsule of shoulder joint.
 - c- Long head of biceps brachii muscle.
 - d- Long head of triceps.
- 8- Regarding the muscles of the upper arm:**
- a- The biceps is a powerful pronator of the forearm.
 - b- The brachialis is supplied by medial and musculocutaneous nerves.
 - c- The biceps is inserted into the ulnar tuberosity.
 - d- The brachialis is the primary flexor of the elbow joint.
- 9- Regarding muscle actions of the upper limb:**
- a- Flexor digitorum superficialis flexes distal phalanges.
 - b- Biceps brachii is a powerful supinator.
 - c- Flexor carpi ulnaris is flexor and abductor of the wrist.
 - d- Pronator muscles are supplied by ulnar nerve.
- 10- Pronator muscles of forearm are supplied by:**
- a- Musculocutaneous nerve
 - b- Radial nerve.
 - c- Ulnar nerve.
 - d- Median nerve.
- 11- The flexor digitorum profundus, choose correct answer:**
- a- Receives double nerve supply.
 - b- Inserted into the front of middle phalanx.
 - c- Flexes the lateral four fingers.
 - d- Arises from radius, ulna and interosseous membrane.
- 12- Mark muscle which receives innervation from 2 different nerves:**
- a- Biceps brachii.
 - b- Pronator teres.
 - c- Flexor digitorum profundus.
 - d- Flexor digitorum superficialis.
- 13- Anterior aspect of interosseous membrane gives attachment to:**
- a- Flexor pollicis longus and flexor digitorum profundus.
 - b- Flexor digitorum superficialis.
 - c- Pronator teres.
 - d- Flexor carpi ulnaris.

- 14- The flexor pollicis longus muscle, choose the correct answer:**
- a- Arises from anterior surface of the ulna and interosseous membrane.
 - b- Is inserted in palmar side of the distal phalanx of thumb.
 - c- Is innervated by branch from posterior interosseous nerve.
 - d- Helps in adduction of wrist joint.
- 15- Pronator quadratus, choose the correct answer:**
- a- Arises from anterior surface of the distal fourth of the radius.
 - b- Inserted into anterior surface of the distal fourth of the ulna.
 - c- Pronates the forearm at radioulnar joints.
 - d- Is innervated by posterior interosseous nerve.
- 16- Regarding the cubital fossa:**
- a- It is bounded laterally by the brachialis.
 - b- It is bounded medially by the pronator teres.
 - c- The basilic vein runs through its roof
 - d- Median nerve lies in the fossa on the lateral side of brachial artery.
- 17- Regarding the structures passing in the roof of the cubital fossa, mark one correct statement:**
- a- Median cubital vein.
 - b- Med-cut-n of forearm.
 - c- Median nerve.
 - d- Ulnar nerve.
- 18- The following structure is a content of the cubital fossa:**
- a- Bracioradialis
 - b- Median nerve.
 - c- Musculocutaneous
 - d- Ulnar nerve.
- 19- The thenar eminence consists of, choose the correct answer:**
- a- Abductor pollicis brevis and adductor pollicis.
 - b- Flexor pollicis longus.
 - c- Abductor pollicis brevis, oppose pollicis, flexor pollicis brevis.
 - d- Flexor pollicis longus, abductor pollicis longus.
- 20- As regards the muscles of the upper limb, one is correct:**
- a- The deltoid is supplied by the axillary nerve.
 - b- All interossei are supplied by the median nerve.
 - c- Dorsal interossei are adductors of the fingers.
 - d- Flexor digitorum superficialis flexes terminal interphalangeal joints.

- 21- The median nerve innervates the following muscles:**
- a-** Extensor carpi radialis longus.
 - b-** Dorsal and palmar interossei.
 - c-** Hypothenar muscles.
 - d-** Lateral two lumbricals.
- 22- The median nerve supplies, one of the following muscle:**
- a-** Flexor carpi ulnaris muscle.
 - b-** Dorsal interossei.
 - c-** Opponens pollicis.
 - d-** Adductor pollicis muscle.
- 23- Regarding the median nerve injury, choose the correct answer:**
- a-** It may be injured by fractures of the shaft of the humerus.
 - b-** The resulting deformity of the hand is called claw hand.
 - c-** There is loss of opposition of the thumb.
 - d-** There is loss of sensation over the lateral 2/3 of dorsum of hand.
- 24- Regarding the effects of median nerve injury, choose the correct answer:**
- a-** The patient can adduct and abduct the fingers.
 - b-** There is an ape's-like deformity of the hand.
 - c-** There is loss of adduction of the thumb.
 - d-** There is loss of sensations on palmar aspect of medial 1½ fingers.
- 25- Regarding the effect of median nerve injury above the elbow, choose the correct answer:**
- a-** Loss of opposition of the little finger.
 - b-** Loss of supination of the forearm.
 - c-** Loss of sensations over the lateral 2/3 of the palm.
 - d-** Weak flexion of the wrist with radial deviation.
- 26- Median nerve entrapment frequently occurs in one of the following points, choose the correct answer:**
- a-** Between the two heads of flexor carpi ulnaris.
 - b-** Deep fascia between flexor digitorum superficialis & flexor pollicis longus.
 - c-** Flexor retinaculum.
 - d-** Palmar aponeurosis.

- 27- Ape's-like hand deformity results from following injury of which one of the following nerves?**
- a- Axillary nerve.
 - b- Median nerve.
 - c- Ulnar nerve.
 - d- Radial nerve.
- 28- A patient suffering from carpal tunnel syndrome, would present with one of the following symptoms:**
- a- Parasthesia over the index finger.
 - b- Parasthesia over the little finger.
 - c- Weakness in adduction of the thumb.
 - d- Weakness in opposition of the little finger.
- 29- Motor loss resulting from carpal tunnel syndrome induces; mark the most acceptable answer:**
- a- Loss of thumb adduction and abduction.
 - b- Loss of writing position of the medial four fingers.
 - c- Loss of opposition of the thumb.
 - d- Loss of abduction and adduction of the medial 4 fingers.
- 30- Regarding the ulnar nerve:**
- a- It is a branch of lateral cord of brachial plexus.
 - b- It gives no branches in forearm.
 - c- It passes behind medial epicondyle of humerus.
 - d- It supplies extensor carpi ulnaris.
- 31- One of the following structures pass behind the medial epicondyle:**
- a- Median nerve.
 - b- Interosseous recurrent artery.
 - c- Musculocutaneous nerve.
 - d- Ulnar nerve
- 32- Regarding the course of the ulnar nerve; mark the correct answer:**
- a- Passes behind the lateral epicondyle of the humerus.
 - b- Passes between humeral and ulnar attachment of flexor carpi ulnaris.
 - c- Passes between flexor digitorum profundus and Palmaris longus.
 - d- Passes between flexor digitorum superficialis & flexor digitorum profundus.

- 33- Which of the following muscle may be paralyzed as a result of fracture of the medial epicondyle of the humerus?**
- a- First lumbrical.
 - b- Extensor digiti minimi.
 - c- Adductor pollicis.
 - d- Flexor carpi radialis.
- 34- Deep branch of the ulnar nerve ends by supplying:**
- a- Abductor digiti minimi.
 - b- Flexor digiti minimi.
 - c- Opponens digiti minimi.
 - d- Adductor pollicis.
- 35- Partial claw hand deformity results from injury of which one of the following nerves?**
- a- Axillary nerve.
 - b- Median nerve.
 - c- Ulnar nerve.
 - d- Radial nerve.
- 36- Loss of finger's abduction and adduction results from lesion of which one of the following nerves?**
- a- Ulnar nerve.
 - b- Median nerve.
 - c- Radial nerve.
 - d- Axillary nerve.
- 37- Regarding the ulnar nerve injury, choose the correct answer:**
- a- The patient can grip a piece of paper placed between the thumb and index.
 - b- There is loss of sensation over the lateral 1/3 of the hand.
 - c- There is loss of sensation over the lateral 3½ fingers.
 - d- If injured above the elbow, the clawing of the hand becomes less apparent than when it is injured above the wrist.
- 38- Regarding nerve injuries in upper limb, choose correct answer:**
- a- Injury to the median produces wrist drop.
 - b- Injury to the ulnar nerve produces ape hand.
 - c- Ulnar nerve is injured by compression in the carpal tunnel.
 - d- Injury to the axillary nerve leads to loss of abduction of the shoulder.

39- The tendon of the following muscle passes superficial to the flexor retinaculum at wrist:

- a- Flexor digitorum superficialis tendons.
- b- Flexor digitorum profundus tendons.
- c- Palmaris longus tendon.
- d- Flexor carpi radialis tendon.

40- The following structure passes superficial to flexor reticulum:

- a- Flexor digitorum superficialis.
- b- Palmar carpal vessels.
- c- Flexor pollicis longus.
- d- Ulnar nerve.

41- The following structure passes deep to the flexor retinaculum:

- a- Ulnar nerve.
- b- Ulnar vessels.
- c- Tendon of palmaris longus.
- d- Median nerve.

42- Regarding synovial sheaths of flexor tendons at the wrist, one is correct:

- a- The radial bursa is the synovial sheath surrounding flexor carpiradialis tendon.
- b- The radial bursa extends for about 2 finger breadth above the flexor retinaculum.
- c- The ulnar bursa is the synovial sheath, which surrounds tendons of flexor digitorum profundus, and superficialis.
- d- Tenosynovitis of the ring finger may extend to the radial bursa.

43- Regarding the thenar space of the hand, choose the correct answer:

- a- It is bounded laterally by the medial septum of the palmar aponeurosis.
- b- It is bounded posteriorly by the fascia covering the adductor pollicis.
- c- It contains the tendon of the digitorum profundus.
- d- It contains the medial three lumbricals.

44- Regarding the midpalmar space of the hand, choose the correct answer:

- a-** It contains the lateral two lumbricals.
- b-** It is bounded medially by the intermediate septum of the palmar aponeurosis.
- c-** It contains the superficial palmar arch.
- d-** It contains the superficial branch of the ulnar nerve.

45- The palmar aponeurosis; choose the correct answer:

- a-** Gives origin to palmaris longus muscle.
- b-** Gives origin to the lumbrical muscles.
- c-** Gives insertion to the adductor pollicis muscle.
- d-** Gives origin to the opponens pollicis muscle.

46- Synovial sheath of digital flexors encloses, mark correct statement:

- a-** Flexor digitorum brevis.
- b-** Flexor digitorum profundus.
- c-** Flexor pollicis brevis.
- d-** Flexor carpi radialis.

47- The palmar thenar space contains, mark the correct answer:

- a-** Flexor pollicis longus tendon.
- b-** Flexor tendons of the middle 3 fingers.
- c-** The 2nd lumbrical muscle.
- d-** Deep palmar arch.

48- Flexor pollicis brevis; choose the correct answer:

- a-** Arises from flexor retinaculum and tubercle of trapezium.
- b-** Lies on the lateral to abductor pollicis brevis.
- c-** Is inserted on the medial side of proximal phalanx (thumb).
- d-** Is innervated by of twig from the superficial branch of ulnar nerve.

49- The opponens pollicis; choose the correct answer:

- a-** Arises from flexor retinaculum and capitate bone.
- b-** Lies superficial to abductor pollicis brevis.
- c-** Is innervated by of twig from ulnar nerve.
- d-** Is inserted into the lateral aspect of the first metacarpal bone.

50- Palmaris brevis; one is correct:

- a- Arises from trapezium and capitate bone.
- b- Is the only muscle innervated by superficial branch of ulnar nerve.
- c- Covers the ulnar artery and the superficial branch of median nerve.
- d- Covers the palmar cutaneous branch of median nerve.

51- The lumbrical muscles, choose the correct answer:

- a- Arise from the tendons of flexor digitorum superficialis.
- b- Are inserted into the radial side of extensor expansion.
- c- Are innervated by ulnar and radial nerves.
- d- In association with the interossei, extend the digits at the metacarpophalangeal joints.

52- The mid palmar space, choose the correct answer:

- a- It contains tendon of flexor pollicis longus.
 - b- It is bounded anteriorly by flexor retinaculum.
 - c- It contains deep palmar arch.
- It contains tendons of the long flexors of medial 3 fingers.

53- The space of Parona, choose the correct answer:

- a- Bounded posteriorly by the pronator quadratus muscle.
- b- Continuous distally with the thenar space.
- c- Continuous distally with the pulp space.
- d- Transmits sepsis to hand.

54- The pulp space, choose the correct answer:

- a- Is present on the palmar aspect of the middle phalanx.
- b- Divided wide spaces by fibrous septae.
- c- Contains the end of the fibrous flexor sheath.
- d- Transmits blood vessels for the shaft and head of the phalanx.

1- B	24- B
2- B	25- C
3- A	26- C
4- B	27- B
5- A	28- A
6- C	29- C
7- A	30- C
8- D	31- D
9- B	32- B
10- D	33- C
11- A	34- D
12- C	35- C
13- A	36- A
14- B	37- D
15- C	38- D
16- B & C	39- C
17- A	40- D
18- B	41- D
19- C	42- C
20- A	43- B
21- D	44- C
22- C	45- D
23- C	46- B
	47- A
	48- A
	49- D
	50- B
	51- B
	52- D
	53- A
	54- D

FILL IN THE BLANKS

- 1- The Coracobrachialis can do and of the shoulder joint.
- 2- The medial half of the brachialis is supplied by the nerve ,while its lateral half is supplied by the nerve.
- 3- The musculocutaneous nerve supplies , and half of
- 4- The flexor carpi ulnaris is inserted into, and the base of
- 5- The synovial sheath of the tendon of the flexor pollicis longus is called bursa. The common synovial sheath of the tendons of the flexor digitorum superficialis and profundus is called bursa.
- 6- The lateral boundary of the cubital fossa is while its medial boundary is
- 7- The floor of the cubital fossa is formed of and muscles.
- 8- All the interossei are supplied bywhile the lateral two lumbricals are supplied by and the medial two lumbricals are supplied by
- 9- The lumbricals can do flexion of joints and extension of joints.
- 10- The palmar interossei can do adduction of, and fingers.
- 11- The dorsal interossei can do abduction of,and fingers.
- 12- The median nerve lies on the side of the upper part of the brachial artery, and on the side of its lower part.
- 13- In the upper 2/3 of the forearm, the median nerve descends between and.....
- 14- In the cubital fossa the median nerve gives muscular branches to,, and
- 15- Carpal tunnel syndrome results from compression of the nerve while passing through the carpal tunnel. The resulting deformity is called
- 16- The nerve and the artery pass behind the medial epicondyle of the humerus.

- 17- In the region of the elbow the ulnar nerve gives muscular branches to and
- 18- In the middle 1/3 of the forearm, the ulnar nerve and artery descend vertically between and muscles. During their course in the forearm, the ulnar nerve descends on the side of the ulnar artery
- 19- Inability of the patient to grip a piece of paper between two fingers is a sign of..... nerve injury, and it is due to paralysis of
- 20- Inability of the patient to grip a piece of paper between the thumb and index finger is a sign of nerve injury, and it is due to paralysis of
- 21- The..... nerve passes between the two heads of the pronator teres, while the..... nerve passes between the two heads of the flexor carpi ulnaris.
- 22- If the median nerve is injured above the elbow, there will be ulnar deviation of the hand due to paralysis of If the ulnar nerve is injured above the elbow, there will be radial deviation of the hand due to paralysis of
- 23- The flexor retinaculum is attached medially to and and laterally to and
- 24- The special tunnel formed by the flexor retinaculum in front of the trapezium transmits the tendon of which is inserted into.....
- 25- The structures that pass superficial to the flexor retinaculum include,,, and
- 26- The structures that pass through the carpal tunnel include,,and.....
- 27- The ulnar artery descends to the flexor retinaculum, while the median nerve passes to the retinaculum.
- 28- In the thumb, the fibrous flexor sheath covers the tendon of, while in each of the medial 4 fingers it covers the tendons of and

- 1- Flexion, adduction.**
- 2- Musculocutaneous, radial.**
- 3- Coracobrachialis, biceps, brachialis.**
- 4- Pisiform, hook of hamate, 5th metacarpal.**
- 5- Radial, ulnar.**
- 6- Brachioradialis, Pronator teres.**
- 7- Brachialis, supinator.**
- 8- Ulnar nerve, median nerve, ulnar nerve.**
- 9- Metacarpophalangeal, interphalangeal.**
- 10- 2nd, 4th, 5th.**
- 11- 2nd, 3rd, 4th.**
- 12- Lateral, medial.**
- 13- Flexor digitorum superficialis, flexor digitorum profundus.**
- 14- Pronator teres, flexor carpi radialis, palmaris longus, flexor digitorum superficialis.**
- 15- Median, ape's hand.**
- 16- Ulnar, superior ulnar collateral.**
- 17- Flexor carpi ulnaris, flexor digitorum profundus.**
- 18- Flexor carpi ulnaris, flexor digitorum profundus, medial.**
- 19- Ulnar, interossei.**
- 20- Ulnar, adductor pollicis.**
- 21- Median, ulnar.**
- 22- Flexor carpi radialis, flexor carpi ulnaris.**
- 23- Pisiform, hamate, scaphoid, trapezium.**
- 24- Flexor carpi radialis, bases of 2nd and 3rd metacarpals.**
- 25- Palmaris longus tendon, ulnar artery, ulnar nerve, palmar cutaneous branch of ulnar nerve, palmar cutaneous branch of median nerve.**
- 26- Tendons of flexor digitorum superficialis, tendons of flexor digitorum profundus, tendon of flexor pollicis longus, median n.**
- 27- Superficial, deep.**
- 28- Flexor pollicis longus, flexor digitorum superficialis, flexor digitorum profundus.**

CROSS MATCHING

Select from column (B) the muscle supplied by the nerve in column (A)

-1-

Column A	Column B
1- Ulnar nerve	a) Brachioradialis
2- Radial nerve	b) Pronator teres
3- Axillary nerve	c) Flexor carpi ulnaris
4- Musculocutaneous nerve	d) Deltoid
5- Median nerve	e) Coracobrachialis

-2-

Select from column (B) the branch of the nerve in column (A):

Column A	Column B
1- Median nerve	a) Posterior interosseous nerve
2- Lateral cord of brachial plexus	b) Axillary nerve
3- Medial cord of brachial plexus	c) Musculocutaneous nerve
4- Radial nerve	d) Ulnar nerve
5- Posterior cord of brachial plexus	e) Anterior interosseous nerve

-3-

Select from column (B) the source of the nerve supply of the muscle in column (A)

Column A	Column B
1- Deltoid	a) Lateral cord of brachial plexus
2- Serratus anterior	b) Posterior cord of brachial plexus
3- Supraspinatus	c) Roots of brachial plexus
4- Coracobrachialis	d) Medial cord of brachial plexus
5- Flexor carpi ulnaris	e) Upper trunk of brachial plexus

-4-

Regarding the muscles of the forearm, select from column (B) what suit the muscle in column (A)

Column A	Column B
1- Flexor carpi radialis	a) May be absent
2- Palmaris longus	b) Median nerve passes between its 2 heads
3- Pronator quadratus	c) Can abduct the hand
4- Pronator teres	d) Supplied by the ulnar nerve
5- Flexor carpi ulnaris	e) Supplied by anterior interosseous nerve

-5-

Regarding the muscles of the upper limb, choose from column (B) what suits the muscle in column (A):

Column A	Column B
1- Biceps	a) Arises from two bones
2- Triceps	b) Arises from three bones
3- Palmaris longus	c) Receives double nerve supply
4- Flexor digitorum superficialis	d) Inserted in the radial tuberosity
5- Brachialis	e) Its tendon is superficial to flexor retinaculum

-6-

Regarding the muscles attached to the scapula, select from column (B) what suit the muscle in column (A)

Column A	Column B
1- Deltoid	a) Upper lip of the spine
2- Long head of biceps	b) Tip of coracoid process
3- Long head of triceps	c) Infraglenoid tubercle
4- Trapezius	d) Lower lip of the spine
5- Coracobrachialis	e) Supraglenoid tubercle

-1-	-2-	-3-	-4-	-5-	-6-
1-c	1-e	1-b	1-c	1-d	1-d
2-a	2-c	2-c	2-a	2-a	2-e
3-d	3-d	3-e	3-e	3-e	3-c
4-e	4-a	4-a	4-b	4-b	4-a
5-b	5-b	5-d	5-d	5-c	5-b

PROBLEM SOLVING

-1-

A house wife complains from numbness and tingling sensation in the hand and lateral fingers which was progressive. On examination, there was some wasting in the thenar eminence.

- 1- What anatomical structure is diseased in this patient?**
- 2- Describe the area of the expected sensory impairment?**
- 3- Does this sensory impairment markedly affect the hand function? Why?**
- 4- What muscles would be functionally impaired?**
- 5- Does this motor impairment markedly affect the hand function? Why?**
- 6- What is the name of this syndrome? What is its usual cause?**
- 7- What is the name of the resulting deformity?**

- 1- Median nerve.**
- 2- Sensory loss over the palmar surface of the lateral 3 1/2 fingers and over the dorsal aspect of the distal phalanx of these fingers.**
- 3- Yes this sensory loss will affect the function of the hand as the sense of touch and proprioception is important for fine manipulations.**
- 4- The thenar muscles and the lateral 2 lumbricals.**
- 5- Loss of opposition of the thumb markedly affects the fine function of the hand and the hand grip.**
- 6- Carpal tunnel syndrome. Compression of the median nerve as it passes deep to the flexor retinaculum.**
- 7- Ape's hand.**

-2-

Following an accident, a middle age man suffered from fracture in the medial epicondyle.

- 1- What structure is liable to be injured?**
- 2- What are the muscles that would be paralyzed?**
- 3- What is the expected sensory loss?**
- 4- What is the name of the resulting deformity?**
- 5- Would this deformity be less severe if the nerve was injured at the wrist? Why?**

- 1-** Ulnar nerve.
- 2-** Flexor carpi ulnaris, medial 1/2 of flexor digitorum profundus, hypothenar muscles, medial 2 lumbricals, dorsal & palmar interossei, adductor pollicis.
- 3-** Sensory loss over the medial 1/3 of the palm and dorsum of the hand and palmar and dorsal aspects of the medial 1 1/2 finger.
- 4-** Claw hand.
- 5-** If the injury was near the wrist the deformity would be less severe by sparing flexor carpi ulnaris and, medial 1/2 of flexor digitorum profundus.

Back of forearm, radial nerve, Joints

M.C.Q'S

- 1- Regarding muscles of the forearm, choose the correct answer:**
 - a- The flexor digitorum profundus arises from three bones.
 - b- The flexor carpi ulnaris abducts the wrist.
 - c- The brachioradialis is innervated by the median nerve.
 - d- The supinator is innervated by the posterior interosseous nerve.
- 2- Mid-prone position of the forearm is achieved by, mark the most appropriate answer:**
 - a- Pronator teres.
 - b- Pronator quadratus.
 - c- Brachioradialis.
 - d- Biceps brachii.
- 3- Supinator muscle; choose the correct answer:**
 - a- Inserted into the upper one third of ulna.
 - b- Surrounds the posterior interosseous artery.
 - c- Is innervated by radial nerve itself.
 - d- Is innervated by posterior interosseous nerve.
- 4- Supination is more powerful than pronation due to; mark the most suitable answer:**
 - a- The action of biceps brachii muscle and supinator.
 - b- The action of supinator muscle only.
 - c- The action of brachioradialis only.
 - d- The action of brachialis.
- 5- Regarding the thumb, choose the correct answer:**
 - a- It is supplied by ulnar artery.
 - b- It is extended by one muscle.
 - c- It is adducted by two muscles.
 - d- It is abducted by two muscles.
- 6- Regarding extensor retinaculum, choose the correct answer:**
 - a- It is attached medially to the pisiform and the hamate.
 - b- It is attached laterally to the anterior border of the lower end of the radius.
 - c- Five compartments for passage of tendons of extensor muscles lie deep to the retinaculum.
 - d- The fourth compartment contains the post-interosseous artery.

- 7- One of following structures lies deep to extensor retinaculum:**
- a- The basilic vein.
 - b- The dorsal cutaneous branch of the ulnar nerve.
 - c- Extensor indices.
 - d- The beginning of the cephalic vein
- 8- Regarding anatomical snuff box the following statements one is wrong:**
- a- It is bounded medially by the tendon of extensor pollicis longus.
 - b- It is bounded laterally by the tendons of abductor pollicis longus and extensor pollicis brevis.
 - c- The radial artery traverses it before entering the palm.
 - d- The capitate can be felt in its floor.
- 9- The radial artery in the anatomical snuff box passes on, choose the correct bone(s):**
- a- Scaphoid.
 - b- Pisiform.
 - c- Trapezoid.
 - d- Capitate.
- 10- The following muscle is innervated by the radial nerve:**
- a- Supinator.
 - b- Abductor pollicis longus.
 - c- Extensor pollicis longus.
 - d- Extensor carpi radialis longus.
- 11- The following muscle is innervated by the radial nerve:**
- a- Long head of biceps brachii.
 - b- Short head of biceps brachii.
 - c- Brachialis.
 - d- Coracobrachialis.
- 12- The radial nerve, around the elbow, choose the correct answer:**
- a- Comes in contact with the medial epicondyle of the humerus.
 - b- Passes between biceps brachii and brachioradialis.
 - c- Pierces the lateral intermuscular septum and passes between brachialis and both brachioradialis and extensor carpi radialis longus.
 - d- Passes between extensor carpi radialis longus and brevis.

- 13- The nerve commonly injured following fracture of the shaft of humerus is:**
- a- Axillary nerve.
 - b- Ulnar nerve.
 - c- Radial nerve.
 - d- Median nerve.
- 14- Wrist drop deformity results from injury of which one of the following nerves?**
- a- Ulnar nerve.
 - b- Median nerve.
 - c- Radial nerve.
 - d- Axillary nerve.
- 15- Regarding radial nerve injury, choose the correct answer:**
- a- It may be injured if the shaft of the humerus is fractured.
 - b- Injury of the posterior interosseous nerve causes wrist drop.
 - c- Injury of its superficial branch produces finger drop.
 - d- Injury of posterior interosseous nerve is accompanied by loss of sensations.
- 16- Regarding injury of radial nerve in the spiral groove, choose the correct answer:**
- a- It results in complete paralysis of triceps muscle.
 - b- It produces paralysis of the extensor carpi ulnaris.
 - c- It results in Ape-like hand deformity.
 - d- It produces disturbed sensation of medial 1/3 of the dorsum of hand.
- 17- Regarding the cutaneous supply of the upper limb, choose the correct answer:**
- a- The skin of the medial side of the forearm is supplied by a branch of the post cord of brachial plexus.
 - b- The skin of the lateral side of the arm is supplied by branches of the lateral cord of the brachial plexus.
 - c- The skin of the dorsum of the hand is supplied by branches of radial and ulnar nerves.
 - d- The skin of the medial 1½ fingers is supplied by branches of the median nerve.

- 18- Cutaneous innervation of the palm, choose the correct answer:**
- a- Ulnar, median, radial and medial cutaneous nerves of forearm.
 - b- Cutaneous branches from ulnar and median nerves.
 - c- Median, ulnar and lateral cutaneous nerves of forearm.
 - d- Ulnar and radial nerve only.
- 19- Regarding lesions of the upper trunk or upper roots of the brachial plexus (Erb's Duchenne palsy), one is correct:**
- a- The shoulder is laterally rotated.
 - b- The forearm is pronated.
 - c- There is loss of sensation on lateral side of arm.
 - d- The elbow is flexed.
- 20- Regarding lesion of the lower root of the brachial plexus, choose the correct answer:**
- a- There is paralysis of the intrinsic muscles of the hand.
 - b- It results in ape hand deformity.
 - c- It can be due to fracture of clavicle.
 - d- There is loss of skin sensations along the lateral side of the arm.
- 21- The clavicle, mark the correct answer:**
- a- Is the only long bone which lies horizontally.
 - b- Contains large medullary cavity.
 - c- The supraclavicular nerves descend behind its middle third.
 - d- The coracoclavicular ligament is attached to the lateral half of the subclavius groove.
- 22- The junction of the middle and lateral thirds of the clavicle is the commonest point of fracture due to; mark the most appropriate statement:**
- a- Change in the contours of the clavicle; circular medially and flattened laterally and junction of two curves.
 - b- The attachment of conoid and trapezoid ligaments.
 - c- The attachment of deltoid and trapezius muscles on anterior and posterior sides of the clavicle.
 - d- Small medullary cavity.

23- Sternoclavicular joint, choose the correct answer:

- a- Is a synovial hinge joint.
- b- Contains a fibrocartilagenous disc.
- c- Between clavicle and sternum at the angle of Lewis.
- d- Supported by coracoclavicular ligament.

24- Regarding shoulder joint, one is correct:

- a- Its entire capsule is attached to surgical neck of the humerus.
- b- Its capsule is anteriorly connected to the subacromial bursa.
- c- Supraspinatus and deltoid muscles produce abduction of the joint.
- d- Tendon of short head of biceps takes origin within capsule of joint.

25- Regarding the shoulder joint:

- a- The subscapular bursa doesn't communicate with the joint cavity.
- b- It is supplied by the axillary and suprascapular nerves.
- c- The capsule is strengthened inferiorly by a strong ligament.
- d- Labrum glenoidale and supraglenoid tubercle are extracapsular.

26- Labrum glenoidale, choose the correct answer:

- a- Is attached to the long head of triceps.
- b- Fibro-cartilaginous rim around glenoid cavity.
- c- It lies extracapsular.
- d- It helps in stability of the shoulder joint.

27- Select most acceptable answer allowing free mobility of shoulder:

- a- Head of humerus (ball) 3:4 times as much as the glenoid fossa (socket).
- b- The presence of 3 glenohumeral ligaments.
- c- The pattern of attachment of the shoulder joint capsule.
- d- The presence of the rotator cuff muscles.

28- The glenohumeral ligaments, choose the correct answer:

- a- Strong ligaments of shoulder joint.
- b- Three in number.
- c- All are attached to the whole margin of glenoid cavity.
- d- Guard against posterior dislocation of the humeral head.

29- Transverse humeral ligament:

- a- Extends between lesser tuberosities and anatomical neck.
- b- Fixes the long head of triceps inside the bicipital groove.
- c- Helps in stability of shoulder joint.
- d- Converts the bicipital groove into a tunnel.

30- Principal muscles which produce abduction of shoulder joint:

- a- Pectoralis major.
- b- Middle fibers of deltoid.
- c- Coracobrachialis.
- d- Biceps brachii.

31- Principal muscle concerned in extension of the shoulder joint:

- a- Anterior fibers of deltoid.
- b- Teres minor.
- c- Latissimus dorsi.
- d- Serratus anterior.

32- Principal muscle concerned in medial rotation of the shoulder:

- a- Pectoralis major.
- b- Deltoid middle fibers.
- c- Supraspinatus.
- d- Teres minor.

33- Medial rotation of the shoulder joint can be performed by:

- a- Posterior fibers of the deltoid.
- b- Infraspinatus.
- c- Teres major.
- d- Supraspinatus.

34- Lateral rotation at the shoulder; is done by:

- a- Deltoid.
- b- Subscapularis.
- c- Serratus anterior.
- d- Supraspinatus.

35- The shoulder joint is innervated by twigs from:

- a- Medial pectoral nerve.
- b- Axillary nerve.
- c- Suprascapular nerve.
- d- Musculocutaneous nerve.

36- Middle third of humerus, choose the correct answer:

- a- Deltoid tuberosity, laterally.
- b- Coracobrachialis insertion laterally.
- c- Spiral groove anteriorly.
- d- Bicipital groove posteriorly.

37- Regarding the humerus, choose the correct answer:

- a- The coronoid fossa lies above the trochlea posteriorly.
- b- The radial fossa lies above the capitulum posteriorly.
- c- The coronoid fossa lies above the trochlea anteriorly.
- d- The capitulum is medial to the trochlea.

38- The carrying angle of extended elbow is due to, mark the correct statement:

- a- The position of the capitulum on the front of the humerus only.
- b- The locking of the olecranon process inside the olecranon fossa.
- c- The wider diameter of medial part of trochlea than its lateral part.
- d- The strain exerted by the capsule of both the elbow and proximal radio-ulnar joints.

39- Medial epicondyle of the humerus, choose the correct answer:

- a- Smaller than the lateral epicondyle.
- b- The musculocutaneous crosses in front of it.
- c- The ulnar nerve passes behind it.
- d- Gives attachment to the common extensor origin.

40- The lateral epicondyle of the humerus, choose the correct answer:

- a- Larger than the medial epicondyle of the humerus.
- b- Gives origin to the common flexor tendon.
- c- Gives attachment to the extensor carpi radialis brevis.
- d- Related to the ulnar recurrent artery.

41- Regarding the elbow joint, choose the correct answer:

- a- It is formed by articulation of the lower end of humerus and the upper ends of radius and ulna.
- b- Its flexion is produced by corobrachialis.
- c- Its capsule is attached to the head of radius.
- d- It is a synovial joint of pivot.

42- The ulnar collateral ligament, choose the correct answer:

- a- Attached to the medial epicondyle of the humerus.
- b- Attached to the medial border of olecranon process of ulna.
- c- Attached to triceps and extensor carpi ulnaris.
- d- The ulnar nerve passes deep to the ulnar collateral ligament.

- 43- The radial collateral ligament, choose the correct answer:**
- a- Attached to the neck of the radius.
 - b- Attached to the annular ligament.
 - c- It is continuous with the capsule of the superior radioulnar joint.
 - d- Prevents abduction of the elbow joint.
- 44- The following nerve doesn't innervate the elbow joint:**
- a- Radial nerve.
 - b- Ulnar nerve.
 - c- Median nerve.
 - d- Axillary nerve.
- 45- Annular ligament, choose the correct answer:**
- a- Forms about 3/4 of the osseofibrous ring around neck of radius.
 - b- Attached to the posterior margin of the radial tuberosity.
 - c- Attached to the head of the radius.
 - d- Its upper border blends with capsule and ligaments of elbow joint.
- 46- The interosseous membrane, choose the correct answer:**
- a- Is attached to the anterior borders of radius and ulna.
 - b- Gives origin to abductor pollicis brevis.
 - c- Gives origin to the deep flexors of the hand.
 - d- Pierced by posterior interosseous vessels.
- 47- The radius is fixed (tethered) to the ulna by:**
- a- Articular disc of the wrist joint.
 - b- Interosseous membrane.
 - c- Radial collateral ligament.
 - d- Ulnar collateral ligament.
- 48- Pronation and supination of the forearm occurs at:**
- a- Elbow joint
 - b- Superior and inferior radio-ulnar joints.
 - c- Shoulder joint.
 - d- Radio-carpal joint.

49- Regarding the movements of pronation and supination, choose the correct answer:

- a- The axis of the movements extends from the neck of radius to the head of ulna.
- b- During these movements the head of the radius rotates around itself within the annular ligament.
- c- During these movements the head of the ulna rotates around the lower end of the radius.
- d- The brachialis puts the forearm in the mid prone position.

50- Supination of the radioulnar joints is done mainly by:

- a- Pronator teres.
- b- Pronator quadratus.
- c- Brachioradialis.
- d- Biceps brachii.

51- Concerning with pronation and supination:

- a- Pronation is more powerful than supination.
- b- Biceps when the elbow is extended.
- c- Brachioradialis puts the forearm in mid-prone position.
- d- The radius is fixed, the ulna is the movable bone.

52- The wrist joint is formed by:

- a- Radius, ulna and proximal row of carpus.
- b- Radius, articular disc and proximal row of carpus.
- c- Radius, capitate, hamate and trapezium.
- d- Ulna capitate trapezium and triquetral.

53- The radius articulates with, choose the correct answer:

- a- Trochlea of the humerus.
- b- Scaphoid and lunate.
- c- Lunate and triquetral.
- d- Trapezium and trapezoid.

54- Choose carpal bones which articulate with radius at the wrist:

- a- Scaphoid and triquetral.
- b- Scaphoid and lunate.
- c- Lunate and trapezium.
- d- Scaphoid and capitate.

55- The range of movement, at the wrist joint, in adduction is more than abduction due to, mark one correct answer:

- a- The distal end of ulnar styloid level is at a more proximal level than that of the radius.
- b- The distal surface of the radius is much longer than that of the ulna.
- c- The distribution of the long flexor and extensor tendons to the medial four fingers.
- d- The pattern of attachment of flexor retinaculum to the carpal bones.

56- Flexion adduction of the wrist can be done by:

- a- Flexor pollicis longus.
- b- Flexor digitorum profundus.
- c- Flexor carpi radialis.
- d- Flexor carpi ulnaris.

57- Regarding bones of the forearm, mark one correct statement:

- a- The radius shares in the formation of three joints.
- b- The ulna shares in the formation of two joints.
- c- The styloid process of the ulna projects downwards more than the styloid process of the radius.
- d- The growing end of the radius is its lower end.

58- Regarding the type of joints of the upper limb, choose the correct answer:

- a- Shoulder polyaxial ball and socket.
- b- Elbow uniaxial pivot.
- c- Radioulnar uniaxial hinge.
- d- Wrist biaxial condylar.

59- Regarding types of joints of upper limb, mark one correct statement:

- a- The superior radioulnar joint is of ball and socket variety
- b- The acromioclavicular joint is of saddle variety.
- c- The elbow joint is of pivot variety.
- d- The wrist joint is of ellipsoid variety.

60- Regarding types of the joints of upper limb, choose correct answer:

- a- The elbow joint is synovial hinge joint.
- b- The carpometacarpal joint of the thumb is synovial pivot.
- c- The radioulnar joints are ball and socket.
- d- The wrist joint is synovial saddle.

61- Regarding movements of joints:

- a- Circumduction can occur at the shoulder girdle.
- b- Pronation can occur at the wrist joints.
- c- Adduction and abduction can occur at the elbow joint.
- d- Flexion and extension can occur at the metacarpophalangeal joints.

62- Muscle(s) causing flexion of each joint; mark wrong account:

- a- Shoulder; is pectoralis minor.
- b- Elbow; is brachioradialis.
- c- Wrist; are flexor digitorum superficialis and profundus.
- d- Distal interphalangeal; flexor digitorum superficialis.

63- Regarding carpal bones, choose the correct answer:

- a- The commonest carpal fracture affects pisiform bone.
- b- Tendon of flexor carpi ulnaris lies in groove on crest of trapezium.
- c- The triquetral bone does not share in the formation of the wrist joint.
- d- The first carpometacarpal joint is a synovial saddle joint.

64- Saddle-shaped (biaxial) carpometacarpal, is joint of which finger:

- a- Thumb.
- b- Index.
- c- Middle.
- d- Ring.

65- Mark the longest of the metacarpal bones:

- a- 1st thumb.
- b- 2nd index.
- c- 3rd middle.
- d- 4th ring.

66- The metacarpophalangeal joints, choose the correct answer:

- a- Between the heads of the metacarpal bones and proximal phalanx.
- b- The articular surfaces are covered by yellow elastic cartilage.
- c- The short flexor muscles induce flexion of the corresponding finger.
- d- Adduction of the thumb occurs at the first metacarpophalangeal joint.

67- Interphalangeal joints of medial four fingers, choose correct answer:

- a- All are fibrous joints.
- b- All are uniaxial (hinge) allow flexion and extension.
- c- Flexion of distal interphalangeal joints by flexor digitorum superficialis.
- d- Flexion of proximal interphalangeal joints by flexor digitorum superficialis only.

1- D	29- D	57- D
2- C	30- B	58- A
3- D	31- C	59- D
4- A	32- A	60- A
5- D	33- C	61- D
6- B	34- A	62- A & D
7- C	35- B & C	63- D
8- D	36- A	64- A
9- A	37- C	65- B
10- D	38- C	66- A
11- D	39- C	67- B
12- C	40- C	
13- C	41- A	
14- C	42- A	
15- A	43- B	
16- B	44- D	
17- C	45- D	
18- C	46- C	
19- B	47- B	
20- A	48- B	
21- A	49- B	
22- A	50- D	
23- B	51- C	
24- C	52- B	
25- B	53- B	
26- B	54- B	
27- A	55- A	
28- B	56- D	

FILL IN THE BLANKS

- 1- The structures that pass superficial to the extensor retinaculum include,,, and
- 2- The first compartment, deep to the extensor retinaculum transmits the tendons of and
- 3- The second compartment, deep to the extensor retinaculum transmits the tendons of and
- 4- The fourth compartment, deep to the extensor retinaculum transmits the tendons of and and the terminals of the artery and nerve.
- 5- The third compartment, deep to the extensor retinaculum transmits the tendon of The fifth compartment transmits the tendon of The sixth compartment transmits the tendon of
- 6- The anatomical snuffbox is bounded laterally by the tendons of and, and medially by the tendon of
- 7- The nerve and the artery run in the spiral groove on the back of the shaft of the humerus.
- 8- The upper lateral cutaneous nerve of the arm is a branch of, while the lower lateral cutaneous nerve of the arm is a branch of
- 9- The superficial radial nerve descends in the upper 2/3 of the forearm deep to muscle on the side of the radial artery.
- 10- The posterior interosseous nerve winds around the lateral aspect ofinside the substance of the muscle.
- 11- The posterior interosseous nerve supplies all the muscles of the back of the forearm except,..... and..... which are supplied by the main trunk of the nerve.
- 12- Injury of the radial nerve mostly occurs in fracture of and results in a deformity called.....
- 13- Posterior interosseous nerve injury may result from fracture..... It is not accompanied by wrist drop due to the intactmuscle.

- 14- Injury of the upper roots of the brachial plexus is calledand is accompanied by loss of sensation on the side of the upper limb.
- 15- Motor manifestation of injury of the lower root of the brachial plexus are mainly due to paralysis of andwhich is called
- 16- The coracoclavicular ligament is formed of an anterolateral part called the part and a posteromedial part called part.
- 17- Protraction of the scapula can be done by and
- 18- Retraction of the scapula can be done by , and
- 19- Elevation of the scapula can be done by and
- 20- Depression of the scapula can be done by and
- 21- Upward rotation of the glenoid cavity can be done by and
- 22- Downward rotation of the glenoid cavity can be done by, and
- 23- The shoulder joint is a synovial joint of variety. The glenoid cavity of the scapula is rather deepened by a rim of fibrocartilage called
- 24- At least two openings are found in the capsule of the shoulder joint; one for..... and the other for
- 25- The ligaments blend with the anterior surface of the capsule of the shoulder joint. The ligament keeps the long head of biceps in position in the spiral groove.
- 26- Flexion of the shoulder joint is performed by,,and
- 27- Extension of the shoulder joint is performed by,,and
- 28- Abduction of the shoulder joint is about for degrees and it is performed by and Further abduction occurs mainly at the joint by the action of and
- 29- Adduction of the shoulder joint is performed by,,and
- 30- Medial rotation of the shoulder joint is performed by,,,and

- 31- Lateral rotation of the shoulder joint is performed by,
.....and
 - 32- The and nerves carry proprioception from the shoulder
joint.
 - 33- The elbow joint is a synovial joint of variety while the wrist
joint is a synovial joint of variety.
 - 34- The medial collateral ligament of the elbow is attached to
and
 - 35- The lateral collateral ligament of the wrist is attached to
and
 - 36- Flexion of the elbow joint can be done by and
 - 37- Extension of the elbow joint can be done by and
 - 38- The carrying angle is about degrees and it disappears when
the elbow joint is fully
 - 39- Movements of pronation and supination occur at the joints.
The axis of supination and pronation movements extend from
the above to the below.
 - 40- The anterior surface of the interosseous membrane gives
attachment to and
 - 41- The posterior surface of the interosseous membrane gives
attachment to,, and
 - 42- The,,, bones form the proximal row of the
carpal bones.
 - 43- The bone is the first carpal bone to ossify, while the is
the last carpal bones to ossify.
 - 44- Abduction of the wrist can be done by , and
 - 45- Adduction of the wrist can be done by and
-

- 1- Cephalic vein, basilic vein, superficial branch of radial nerve,
dorsal cutaneous branch of ulnar nerve.
- 2- Abductor pollicis longus, extensor pollicis brevis.
- 3- Extensor carpi radialis longus, extensor carpi radialis brevis.
- 4- Extensor digitorum, extensor indicis, anterior interosseous,
posterior interosseous.

- 5- Extensor pollicis longus, extensor digiti minimi, extensor carpi ulnaris.
- 6- Abductor pollicis longus, extensor pollicis brevis, extensor pollicis longus.
- 7- Radial, profunda brachii.
- 8- Posterior division of axillary nerve, radial.
- 9- Brachioradialis, lateral.
- 10- Shaft of the radius, supinator.
- 11- Brachioradialis, extensor carpi radialis longus, anconeus, radial.
- 12- Middle of the shaft of the humerus, wrist drop.
- 13- Upper part of the shaft of the radius, extensor carpi radialis longus.
- 14- Erb's palsy, lateral.
- 15- Lumbricals, interossei, Klumpke's paralysis.
- 16- Trapezoid, conoid.
- 17- Serratus anterior, pectoralis minor.
- 18- Middle fibers of trapezius, rhomboid major, rhomboid minor.
- 19- Upper fibers of trapezius, levator scapulae.
- 20- Lower fibers of trapezius, pectoralis minor.
- 21- Upper and lower fibers of trapezius, serratus anterior.
- 22- Levator scapulae, rhomboid major, rhomboid minor.
- 23- Ball and socket, labrum glenoidale.
- 24- Tendon of long head of biceps, connection with subscapular bursa.
- 25- Anterior glenohumeral, transverse humeral.
- 26- Clavicular head of pectoralis major, anterior fibers of deltoid, biceps, coracobrachialis.
- 27- Latissimus dorsi, posterior fibers of deltoid, teres major.
- 28- 90, supraspinatus, deltoid, sternoclavicular, upper and lower fibers of trapezius, serratus anterior.
- 29- Pectoralis major, latissimus dorsi, teres major, subscapularis.
- 30- Subscapularis, pectoralis major, latissimus dorsi, teres major, anterior fibers of deltoid.
- 31- Infraspinatus, teres minor, posterior fibers of deltoid.

- 32- Suprascapular, axillary.
- 33- Hinge, ellipsoid.
- 34- Medial epicondyle of humerus, medial side of trochlear notch of ulna.
- 35- Lateral epicondyle of humerus, annular ligament.
- 36- Biceps, brachialis.
- 37- Triceps, anconeus.
- 38- 15, flexed.
- 39- Radioulnar, head of radius, head of ulna.
- 40- Flexor pollicis longus, flexor digitorum profundus.
- 41- Abductor pollicis longus, extensor pollicis brevis, extensor pollicis longus, extensor indicis.
- 42- Scaphoid, lunate, triquetral, pisiform.
- 43- Capitate, pisiform.
- 44- Flexor carpi radialis, extensor carpi radialis longus, extensor carpi radialis brevis.
- 45- Flexor carpi ulnaris, extensor carpi ulnaris.

CROSS MATCHING

-1-

Select from column (B) the nerve producing the movement in column (A):

Column A	Column B
1- Flexion of the elbow joint	a) Radial nerve
2- Extension of the elbow joint	b) Musculocutaneous nerve
3- Pronation at the radioulnar joints	c) Ulnar nerve
4- Abduction of the metacarpophalangeal joints	d) Median nerve
5- Initiation of abduction of the shoulder	e) Suprascapular nerve

-2-

Choose from column (B) the joint which allow the movement in column (A)

Column A	Column B
1- Circumduction	a) Shoulder joint
2- Pronation	b) Radioulnar joints
3-Adduction, abduction, flexion and extension	c) Elbow joint
4- Extension and flexion only	d) Metacarpophalangeal joints.
5- Gliding	e) Acromioclavicular joint

-3-

Select from column (B) the nerve whose injury causes the deformity in column (A):

Column A	Column B
1- Ape's hand	a) Long thoracic nerve
2- Flat shoulder	b) Radial nerve
3- Wrist drop and fingers drop	c) Median nerve
4- Winging of scapula	d) Posterior interosseous nerve
5- Fingers drop with no wrist drop	e) Axillary nerve

-4-

Regarding the nerve supply of the muscles of the forearm, select from column (B) the nerve matching with the muscle in column (A)

Column A	Column B
1- Flexor pollicis longus	a) Median nerve
2- Extensor carpi ulnaris	b) Ulnar nerve
3- Brachioradialis	c) Radial nerve
4- Flexor carpi ulnaris	d) Anterior interosseous nerve
5- Flexor carpi radialis	e) Posterior interosseous nerve

-5-

Select from column (B) the type of each joint in column (A)

Column A	Column B
1- Shoulder joint	a) Hinge
2- Elbow joint	b) Ellipsoid
3- Radioulnar joints	c) Ball and socket
4- Wrist joint	d) Saddle
5- Sternoclavicular joint	e) Pivot

-1-	-2-	-3-	-4-	-5-
1-b	1-a	1-c	1-d	1-c
2-a	2-b	2-e	2-e	2-a
3-d	3-d	3-b	3-c	3-e
4-c	4-c	4-a	4-b	4-b
5-e	5-e	5-d	5-a	5-d

PROBLEM SOLVING

-1-

A 12 years old boy suffered from fracture middle of the shaft of the humerus following a car accident.

- 1- What nerve is liable to be injured?
- 2- What movements would be affected following paralysis of these muscles?
- 3- What is the name of the resulting deformity?
- 4- Why this deformity is considered functionally disabling?
- 5- Mention the site of the sensory loss that would occur?

- 1- The radial nerve.
- 2- Weak extension of the elbow and loss of extension of the wrist & fingers.
- 3- Wrist drop.
- 4- This deformity affects the power of the hand grip as wrist extensors act as synergists for the long flexors of the fingers during power grip of the hand.
- 5- In the skin of the dorsum of the hand over the area between the first and second metacarpal bones, in the skin of back of forearm, skin of back of arm & lower half of the lateral surface of the arm.

-2-

A 50 years old woman fell on her outstretched hand. She had a fracture of the upper third of the shaft of the radius.

- 1- Why in this type of fracture, segments of the fractured bone tend to separate?
- 2- What nerve is liable to be injured?
- 3- What would be the paralyzed muscles?
- 4- What movements would be affected by such paralysis?
- 5- What is the name of the resulting deformity?

- 1- The proximal segment is supinated by supinator and biceps while the distal segment is pronated by pronator teres and pronator quadratus.
- 2- Posterior interosseous nerve.
- 3- All muscles of the back of forearm except brachioradialis, extensor carpi radialis longus, anconeus, supinator & extensor carpi radialis brevis.
- 4- Weak extension of the wrist with radial deviation, loss of extension of the metacarpophalangeal joints of the fingers.
- 5- Finger drop.

-3-

During an obstructed labor, the obstetrician had to pull on the baby's head. The baby was noticed to be unable to move his arm and forearm properly.

- 1- What nervous structure do you expect to be injured?**
- 2- What is the name of this injury?**
- 3- Name 5 muscles that would be paralyzed?**
- 4- What position will the arm and forearm take due to this paralysis?**
- 5- Where do you expect this baby to suffer from skin sensory loss?**

- 1- Upper trunk of the brachial plexus (C5-6).**
- 2- Erb's palsy.**
- 3- Deltoid and teres minor (axillary nerve), subscapularis ant teres major (upper and lower subscapular nerves), supraspinatus and infraspinatus (suprascapular nerve), biceps and brachialis and coracobrachialis (musculocutaneous nerve) [any 5 muscles].**
- 4- The arm is adducted, the forearm is extended and pronated.**
- 5- Along the lateral aspect of the arm and forearm.**

-4-

A rock climbing student fell suddenly and grasped a tree just before reaching the ground with his outstretched hands. He was found to have motor and sensory losses.

- 1- What nervous structure do you expect to be injured?**
- 2- What is the name of this injury?**
- 3- What are the paralyzed muscles?**
- 4- What is the expected deformity?**
- 5- Where would you test for skin sensations in such an injury?**

- 1- Avulsion of the lower trunk of the brachial plexus (C8 & T1).**
- 2- Klumpke's paralysis.**
- 3- Mainly the small muscles of the hand.**
- 4- Complete claw hand.**
- 5- Along the medial aspect of the arm, forearm and hand.**

-5-

A 16 years old girl fell on her outstretched hand and felt sudden pain in her right shoulder. On examination her shoulder was depressed.

- 1-** Which bone of the upper limb do you expect to be fractured?
- 2-** Which part of this bone is most commonly fractured? Why?
- 3-** Why is the shoulder region depressed?
- 4-** What structures are liable to be damaged in such a fracture?
- 5-** What is the procedure usually used for the fixation of such a fracture?

- 1-** The clavicle.
- 2-** The junction between the lateral and middle thirds as it is the site of change in curvature (lateral 1/3 convex posteriorly and medial 2/3 convex anteriorly).
- 3-** The shoulder drops because of the weight of the upper limb which is suspended through coracoclavicular ligament to lateral 1/3 of the clavicle.
- 4-** Subclavian vessels and divisions of the brachial plexus.
- 5-** An arm sling to support the sagging limb.

-6-

An elderly man complains from severe pain with abduction of his right shoulder. Physician diagnosed the case as subacromial bursitis.

- 1-** What is the subacromial bursa?
 - 2-** What tendon does it protect?
 - 3-** What is the nerve supply of the muscle?
 - 4-** Why does pain accompany abduction of the shoulder joint?
 - 5-** What is the action of this muscle?
-
- 1-** It is a closed synovial sac between deltoid muscle & capsule of shoulder joint, it extends between coracoacromial arch & supraspinatus tendon.
 - 2-** Supraspinatus tendon.
 - 3-** Suprascapular nerve.
 - 4-** The swollen and inflamed bursa is compressed by the contracting deltoid and supraspinatus muscles during abduction.
 - 5-** Stability and abduction of the shoulder joint from 0 to 15 degrees.

A 30 years old woman fell on her outstretched hand. She suffered from severe pain in the lateral part of the wrist particularly at the base of the anatomical snuff box.

- 1- What is the expected diagnosis?**
- 2- What are the boundaries of the anatomical snuff box?**
- 3- What are the contents of the anatomical snuff box?**
- 4- What makes the floor of the anatomical snuff box?**
- 5- What makes the roof of the anatomical snuff box?**

- 1- Fracture of scaphoid bone.**
- 2- Medially tendon of extensor pollicis longus, laterally tendons of abductor pollicis longus and extensor pollicis brevis.**
- 3- Radial artery, tendons of extensor carpi radialis longus and brevis.**
- 4- Styloid process of radius, scaphoid and trapezium.**
- 5- Skin superficial fascia containing superficial radial nerve and beginning of cephalic vein.**

Book MCQ's

(upper limb)

UPPER LIMB

1- Which of the following bones is a flat bone:

- a. Humerus
- b. Scapula
- c. Ulna
- d. clavicle

2- One of the following bones undergoes membranous ossification:

- a. Clavicle
- b. Radius
- c. Humerus
- d. Ulna

3-The following muscle has double innervation:

- a) Pectoralis major
- b) Pectoralis minor
- c) Subclavius
- d) Deltoid

x 4-The axillary tail of the breast is related to:

- a) Pectoral group of lymph nodes.
- b) Apical group of lymph nodes
- c) Central group of lymph nodes.
- d) Humeral group of lymph nodes.

5-Paralysis of the following muscle leads to shoulder drop:

- a) Pectoralis major
- b) Latissimus dorsi
- c) trapezius
- d) deltoid

6-The dorsal scapular nerve arises from:

- a) Posterior cord of brachial plexus
- b) Upper trunk of brachial plexus
- c) Roots of brachial plexus
- d) Medial cord of brachial plexus

7-Paralysis of the following muscle leads to flat shoulder:

- a) Pectoralis major
- b) Latissimus dorsi
- c) Trapezius
- d) Deltoid

8-Winging of the scapula is due to:

- a) Paralysis of serratus anterior
- b) Paralysis of pectoralis major
- c) Paralysis of deltoid
- d) Paralysis of teres major

9-Choose the correct answer, which of the following muscles forms the anterior fold of axilla:

- a. Deltoid muscle
- b. Pectoralis major
- c. Pectoralis minor
- d. Subclavius

10-The continuation of musculocutaneous nerve is:

- a. Posterior cutaneous nerve of forearm
- b. Medial cutaneous nerve of forearm
- c. Lateral cutaneous nerve of forearm
- d. Medial cutaneous nerve of arm

11-Concerning the boundaries of the cubital fossa:

- a. The floor is formed by the brachialis and supinator muscles.
- b. The roof is formed by the skin, fascia and bicipital aponeurosis.
- c. Lateral boundary is formed by lateral margin of brachioradialis muscle.
- d. Medial boundary is formed by medial margin of pronator teres muscle.

12-Concerning the contents of the cubital fossa, one is wrong:

- a. Median nerve.
- b. Termination of the radial artery.
- c. Ulnar artery.
- d. Radial nerve.

13- Regarding anatomical snuff box, one statement is incorrect:

- a. It is bounded medially by the tendon of extensor pollicis longus.
- b. It is bounded laterally by the tendons of abductor pollicis longus and extensor pollicis brevis.
- c. The cephalic vein usually crosses its roof.
- d. The capitate can be felt in its floor.

14-The Interosseous muscles of the hand are supplied by:

- a. Deep branch of ulnar nerve
- b. Superficial branch of ulnar nerve
- c. Median nerve
- d. Posterior Interosseous nerve

15-Loss of finger's abduction and adduction results from lesion of which one of the following nerves?

- a. Ulnar nerve.
- b. Median nerve.
- c. Radial nerve.
- d. Musculocutaneous nerve.

16-Winging of scapula follows paralysis of:

- a. Serratus anterior muscle.
- b. Deltoid muscle.
- c. Trapezius muscle.
- d. Teres major muscle.

17-Ape's-like hand deformity results from following injury of which one of the following nerves?

- a. Axillary nerve.
- b. Median nerve.
- c. Ulnar nerve.
- d. Radial nerve.

18-Wrist drop deformity results from injury of which one of the following nerves?

- a. Ulnar nerve.
- b. Median nerve.
- c. Radial nerve.
- d. Axillary nerve.

19-Inability to abduct arm from 15 to 90 degrees is due to injury of:

- a. Radial nerve.
- b. Long thoracic nerve (nerve to serratus anterior).
- c. Musculocutaneous nerve.
- d. Axillary nerve.

20-Principal muscle concerned in lateral rotation of the shoulder:

- a. Pectoralis major.
- b. Deltoid anterior fibers.
- c. Teres major.
- d. Teres minor.

21-The ulnar collateral ligament:

- a. Attached to the medial epicondyle of the humerus.
- b. Attached to the lateral border of coronoid process of ulna.
- c. Attached to triceps and flexor carpi radialis.
- d. The ulnar nerve passes in front the ulnar collateral ligament.

22- Concerning pronation and supination:

- a. Pronation is more powerful than supination.
- b. Supination is done by biceps when the elbow is extended.
- c. They occur at the elbow joint.
- d. The ulna is fixed; the radius is the movable bone.

1-b	2-a	3-a	4-a	5-c	6-c	7-d	8-a	9-b	10-c
11-a & b	12-b	13-d	14-a	15-a	16-a	17-b	18-c	19-d	20-d
21-a	22-d								

Arteries of upper limb

M.C.Q'S

1-One of the followings is a branch of axillary artery:

- a- Profunda brachii artery.
- b- Subscapular artery.
- c- Superior ulnar collateral.
- d- Inferior ulnar collateral.

2-Regarding the ulnar artery, choose the correct answer:

- a- It begins opposite the head of radius.
- b- It ends by forming the deep palmar arch.
- c- Its pulsation can be felt lateral to the flexor carpi ulnaris.
- d- It gives the common interosseous artery.

3-Regarding the axillary artery, choose the correct answer:

- a- It begins at the inner border of the first rib.
- b- It ends at the lower border of the teres minor muscle.
- c- It is divided into three parts by the pectoralis minor muscle.
- d- Its third part is related medially to the median nerve.

4-Regarding radial and ulnar arteries, choose the correct answer:

- a- They arise from brachial artery at the level of the head of the radius.
- b- Both arteries give anterior and posterior carpal branches.
- c- Radial artery forms the superficial palmar arch.
- d- Ulnar artery runs deeply in anatomical snuffbox.

5-Regarding the radial artery, one is correct:

- a- Its pulsations can be felt between tendons of flexor carpi radialis & flexor digitorum superficialis.
- b- It continues in the palm as the superficial palmar arch.
- c- It begins at the level of the head of the radius.
- d- In the forearm, the superficial radial nerve lies medial to the artery.

6-Regarding the axillary artery, one is correct:

- a- The largest of its branches is the superior thoracic artery.
- b- Its third part is crossed by the lateral root of the median nerve.
- c- The pectoralis minor covers most of the artery.
- d- The axillary vein lies medial to the artery.

7-The following artery is a branch of radial artery:

- a- Anterior ulnar recurrent artery.
- b- Posterior ulnar recurrent artery.
- c- Radialis indicis artery.
- d- Common interosseous artery.

8-Regarding the ulnar artery, one is correct:

- a- It descends vertically on medial side of forearm lateral to the ulnar nerve.
- b- It enters the hand by passing deep to the flexor retinaculum.
- c- It continues in the hand as the superficial palmar arch.
- d- It supplies the thumb.

9-The following artery is a branch ulnar artery:

- a- Radialis indicis.
- b- Princeps pollicis.
- c- Superficial palmar branch.
- d- Anterior carpal artery.

10-Regarding the brachial artery, choose the correct answer:

- a- The median nerve lies on the medial of its upper half.
- b- It lies lateral to tendon of brachialis.
- c- It is separated from median cubital vein by the bicipital aponeurosis.
- d- It begins at the lower border of the teres minor.

11-Radial artery in anatomical snuff box passes on, choose correct bone(s):

- a- Scaphoid.
- b- Pisiform.
- c- Trapezoid.
- d- Capitate.

12- The radial artery gives the following branch in the hand:

- a- Principes pollicis.
- b- Superficial palmar artery.
- c- Second dorsal metacarpal artery.
- d- Posterior carpal artery.

13-Branches of ulnar artery in the forearm, choose the correct answer:

- a- Anterior ulnar recurrent.
- b- Ulnar collateral artery.
- c- Superficial palmar branch.
- d- Nutrient branch to humerus.

14-Regarding ulnar artery at wrist & hand, mark the correct answer:

- a- Gives a deep palmar branch.
- b- Gives radialis indicis artery.
- c- Continues as anterior carpal arch.
- d- Continues as deep palmar arch.

15-The brachial artery, choose the correct answer:

- a-** Starts at the distal border of teres major muscle.
- b-** Ends opposite the head of radius.
- c-** Descends on the medial side of the humerus.
- d-** It descends medial to ulnar nerve and bicipital aponeurosis.

16-Profunda brachii artery, choose the correct answer:

- a-** Arises from third part of axillary artery.
- b-** Passes with the ulnar nerve in the spiral groove.
- c-** Descends between medial and long heads of triceps.
- d-** It supplies biceps brachii muscle.

1- B
2- D
3- C
4- B
5- A
6- D
7- C
8- C
9- D
10- C
11- A
12- A
13- A
14- A
15- A
16- C

FILL IN THE BLANKS

- 1- The axillary artery begins at the outer border of the and ends at the lower border of the
- 2- The brachial artery begins at the lower border of as the continuation of
- 3- The brachial artery ends in the cubital fossa at the level of the neck of the by dividing into and arteries.
- 4- The ulnar artery continues in the hand as thepalmar arch, while the radial artery continues as the palmar arch.

- 1- 1st rib, teres major muscle.
- 2- Teres major muscle, axillary.
- 3- Radius, radial, ulnar.
- 4- Superficial, deep.

CROSS MATCHING

Select from column (B) the branche of the artery in column (A)

Column A		Column B		
1- Radial artery		a) Subscapular artery		
2- Ulnar artery		b) Suprascapular artery		
3- Brachial artery		c) Princeps pollicis		
4- Axillary artery		d) Superior ulnar collateral		
5- Subclavian artery		e) Common interosseous		
1-c	2-e	3-d	4-a	5-b

PROBLEM SOLVING

-1-

A young student tried to commit suicide by cutting in front of his wrist with a blade.

- 1- What vessels would be cut?
- 2- Describe 2 points where you can apply pressure to stop the bleeding.
- 3- In what direction should the pressure be applied?
- 4- What nerves would be injured?
- 5- What is the expected neurological loss?

1- Radial and ulnar arteries.

2- The brachial artery can be compressed against the humerus at the medial side of the middle of the arm and can be compressed against the humerus at the lower part of the arm.

3- Backwards.

4- Ulnar nerve, median nerve, palmar cutaneous branch of ulnar nerve, palmar cutaneous branch of median nerve.

5- Loss of sensation over the palmar aspect of the hand and fingers. Motor loss involves all small muscles of the hand.

-2-

A 6 years old child had a pin prick in the tip of her thumb. Two days later, she suffered from severe pain in the anterior surface of the terminal phalanx of the thumb which was swollen and red.

1- What is the possible diagnosis?

2- What is the danger from such infection?

3- What is the lymphatic drainage of this area?

4- Mention the artery supplying the thumb?

5- Mention the nerves supplying the skin of thumb?

1- Pulp space infection.

2- The pus compresses the vessels supplying the terminal part of the distal phalanx leading to necrosis.

3- Lateral group of axillary lymph nodes.

4- Radial artery.

5- Median nerve and superficial radial nerve.

-3-

In the clinic, a sphygmomanometer and a stethoscope are usually used to measure the blood pressure.

1- Where the sphygmomanometer cuff is usually applied?

2- What artery does the inflated cuff compress?

3- Against what bone does it compress the artery?

4- Where is the stethoscope applied?

5- Mention the relations of the artery in this position?

-
- 1-** Around the arm; The inflated cuff should compress the medial side of the 4 arm.
 - 2-** The brachial artery.
 - 3-** The humerus.
 - 4-** On the front of the elbow in the cubital fossa.
 - 5-** It lies between the median nerve medially and the bicipital tendon laterally. It lies deep to the bicipital tendon and superficial to the brachialis muscle at the floor of the cubital fossa.